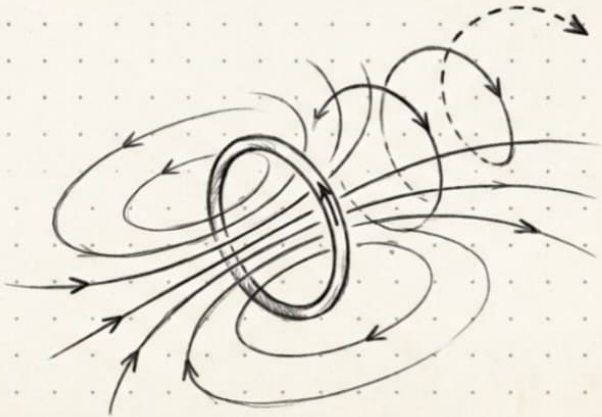
Chapter 4

MOVING CHARGES AND MAGNETISM



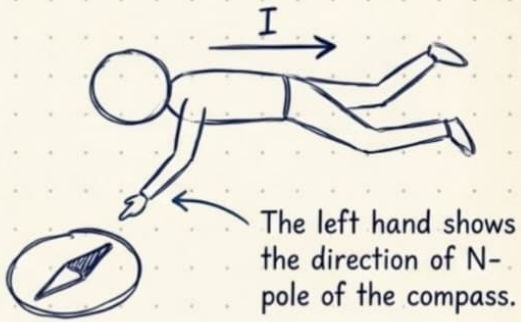
Magnetic Phenomena

1. Magnetic phenomena are a result of forces between charges in motion.
   (i) A moving charge or flowing current in a wire sets up its magnetic field.
   (ii) In a pre-existing magnetic field, if a moving charge or a current-carrying wire is placed in a particular order, it can experience a force due to it.

2. In 1820, a Dutch physicist gave us experimental proof of the fact that a current produces a magnetic field in its surrounding, namely Hans Christian Oersted.

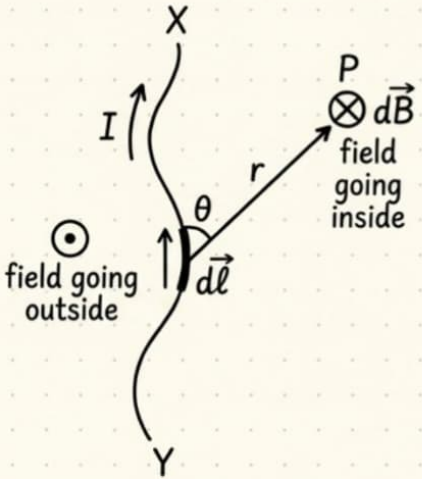
Ampere's Swimming Rule

The direction in which the compass needle will deflect due to a magnetic field produced by a current flowing in a conducting wire is similar to a man swimming in a pool. The direction of the man's head represents the direction of current, and the movement of his left hand shows the direction of deflection.



Biot-Savart Law

Given by 2 French physicists: Jean-Baptiste Biot & Felix Savart.



When $\theta = 0$ (Point P is on the wire), B = 0.
When $\theta = 90$ degrees, B is maximum.

Consider an element $d\ell$ of conductor XY having current I.
Let P be a point at distance r from $d\ell$ where magnetic field dB is to be calculated.

$$dB \propto I,\ d\ell,\ \sin(\theta),\ \text{and}\ \frac{1}{r^2}$$

Combining: $$dB \propto \frac{I \cdot d\ell \cdot \sin(\theta)}{r^2}$$

$$dB = \frac{\mu_0}{4\pi}\frac{I \cdot d\ell \cdot \sin(\theta)}{r^2}$$

Scalar form: $$dB = \frac{\mu_0}{4\pi}\frac{I \cdot d\ell \cdot \sin(\theta)}{r^2}$$

Vector form: $$d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{\ell} \times \vec{r}}{r^3}$$

CGS System:
$1G = 10^{-4}$ T,
$1T = 10^{4}$ G

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$$c^2 = \frac{1}{\mu_0 \varepsilon_0} = (\mu_0 \varepsilon_0)^{-1} = 9 \cdot 10^{16}$$

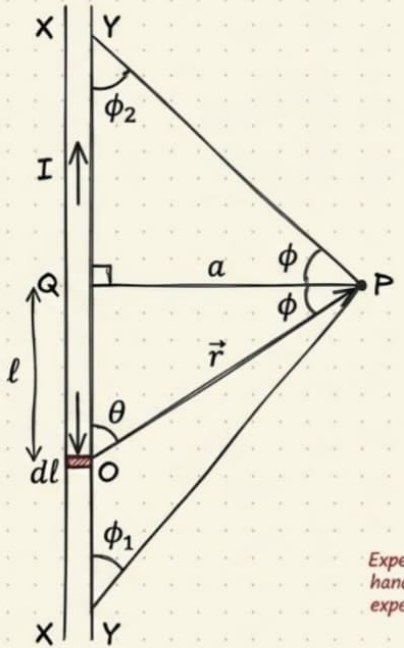
Dimensions of $(\mu_0 \varepsilon_0)^{-1}$
$= [L^2 T^{-2}]$

The speed of light binds electricity and magnetism together!

μ_0 is permeability of free space. Value: $4\pi \cdot 10^{-7}$

Constant of proportionality $\left(\frac{\mu_0}{4\pi}\right) = 10^{-7}$ T/Gauss.

Magnetic Field Due to a Straight Conductor (Part 1: Setup)



Box 1 (The Element)

Consider conductor XY carrying current I. Point P is at perpendicular distance 'a' from XY. Let $d\vec{l}$ be at point O, distance r from P.

Using Biot-Savart: $d\vec{B} = \frac{\mu_0 \cdot I}{4\pi} \cdot \frac{d\vec{l} \cdot \sin(\theta)}{r^2}$ ... (Eq 1)

Expert clarifications hand the mx hand cancellation

Box 2 (Angle Substitution)

In triangle PQO, $\angle QPO + \angle POQ + \angle OQP = 180°$

$\phi + \theta + 90° = 180° \Rightarrow \theta = 90° - \phi$

$\sin(\theta) = \sin(90° - \phi) \Rightarrow \sin(\theta) = \cos(\phi)$ ... (Eq 2)

Expert clarifications and arrows in show it in cancellation

Box 3 (Distance Substitutions)

$\tan(\phi) = \frac{OQ}{PQ} = \frac{l}{a} \Rightarrow l = a * \tan(\phi)$

Differentiating w.r.t ϕ:

$\frac{dl}{d(\phi)} = a * \sec^2(\phi)$. $dl = a * \cancel{\sec^2(\phi)} * \cancel{d(\phi)}$ ... (Eq 3)

$\cos(\phi) = \frac{a}{r} \Rightarrow r = \frac{a}{\cos(\phi)} \Rightarrow r = a * \sec(\phi)$ ... (Eq 4)

Expert clarifications hand on arme hand experts in the wires

Key Insight: Expressing dl and r in terms of the single variable ϕ simplifies the integral!

Laureate's Notebook

Magnetic Field Due to a Straight Conductor (Part 2: Integration & Cases)

Box 4 (Substituting into Eq 1)

$$dB = \frac{\mu_0 I}{4\pi} \cdot \frac{\cancel{a} \cdot \sec^2\phi \cdot d\phi \cdot \cos\phi}{(\cancel{a} \cdot \sec\phi))^2}$$

Cancel common terms

Simplifies expression

$$dB = \frac{\mu_0 I}{4\pi} \cdot \frac{d\phi \cdot \cos\phi}{a}$$

Box 5 (Integration)

Integrating both sides from $-\phi_1$ to ϕ_2:

$$B = \int_{-\phi_1}^{\phi_2} \frac{\mu_0 I}{4\pi a} \cos\phi \cdot d\phi$$

$$B = \frac{\mu_0 I}{4\pi a} [\sin(\phi)]_{-\phi_1}^{\phi_2}$$

$$\boxed{B = \frac{\mu_0 I}{4\pi a} [\sin(\phi_2) + \sin(\phi_1)]}$$

Expert clarifications with simplifies expression

1. Infinitely Long Wire

$\phi_1 = 90°, \phi_2 = 90°$

Expert clarifications in the conducition

$$B_\infty = \frac{\mu_0 I}{4\pi a} [\sin(90) + \sin(90)]$$

$$B_\infty = \frac{\mu_0 I}{4\pi a} [1 + 1] = 2\frac{\mu_0 I}{4\pi a}$$

$$B_\infty = 2 \cdot 10^{-7} \left(\frac{I}{a}\right)$$

2. Semi-Infinitely Long Wire

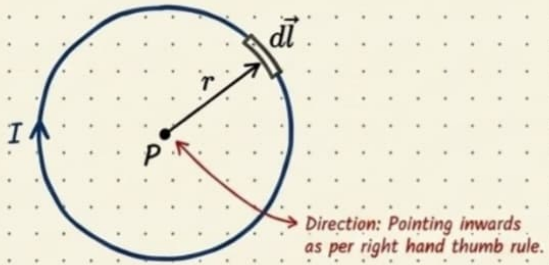
$\phi_1 = 0°, \phi_2 = 90°$

$$B_{\text{semi-Infinity}} = \frac{\mu_0 I}{4\pi a} [\sin(90) + \sin(0)]$$

$$B_{\text{semi-Infinitly}} = \frac{\mu_0 I}{4\pi a} [1]$$

$$B_{\text{semi-Infinitly}} = \frac{\mu_0 I}{4\pi a}$$

Magnetic Field Due to Current Loop & Arcs



$$dB = \frac{\mu_0 \cdot I}{4\pi} \cdot \frac{d\vec{l} \times \vec{r}}{r^3} \Rightarrow dB = \frac{\mu_0 \cdot I}{4\pi} \cdot \frac{d\vec{l} \cdot \sin(\theta)}{r^2}$$

Here, $d\vec{l}$ is perpendicular to $r \Rightarrow \theta = 90°$, $\sin(90°) = 1$.

$$dB = \frac{\mu_0 \cdot I \cdot dl}{4\pi \cdot r^2}$$

Substitute dl

Integrating: $B = \frac{\mu_0 \cdot I}{4\pi r^2} r^2 \int dl$

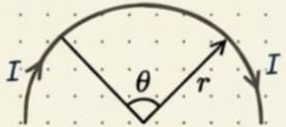
Since $\int dl = 2\pi r$, $B_{center} = \frac{\mu_0 \cdot I \cdot 2\pi r}{4\pi \cdot r^2}$

Simplify

$$\boxed{B_{center} = \frac{\mu_0 \cdot I}{2r}}$$

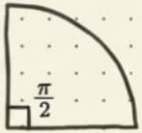
Special Cases

1. Field due to an arc:



$$B = \frac{\mu_0 \cdot I}{4\pi r^2} r^2 (\theta \cdot r) = \frac{\mu_0 \cdot I \cdot \theta}{4\pi \cdot r}$$

2. Field due to a quarter loop:

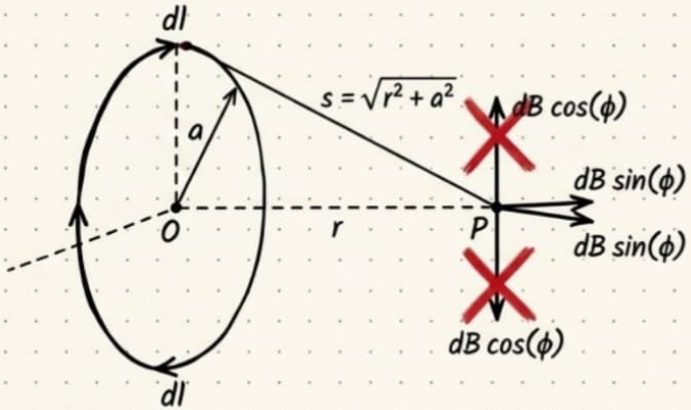


$$\theta = \frac{\pi}{2}$$

$$B = \frac{\mu_0 \cdot I}{4\pi \cdot r} r \left(\frac{\pi}{2}\right) = \frac{\mu_0 \cdot I}{8r}$$

Magnetic Field at the Axis of a Loop

The Cancellation Diagram



This field has 2 components ($dB\cos(\phi)$ and $dB\sin(\phi)$). For a diametrically opposite current element, there are 2 more components. The perpendicular components ($dB\cos(\phi)$) will cancel out! The field will only be because of parallel components ($dB\sin(\phi)$).

Using Biot-Savart: $dB = \dfrac{\mu_0 I \cdot dl}{4\pi \cdot s^2}$

Apply integration limits for full loop

$$B = \int dB \cdot \sin(\phi = \int \left(\frac{\mu_0 I \cdot dl}{4\pi \cdot s^2} \cdot \frac{a}{s} \right)$$

since $\sin(\phi = \frac{a}{s}$

$$B = \frac{\mu_0 I \cdot a}{4\pi \cdot s^3} \int dl$$

$$B = \frac{\mu_0 I a \cdot 2\pi a}{4\pi \cdot s^3} = \frac{\mu_0 I a^2}{2 \cdot s^3}$$

Substitute $s = (r^2 + a^2)^{1/2}$: → $\boxed{B = \dfrac{\mu_0 I a^2}{2(r^2 + a^2)^{3/2}}}$

For N turns: $B = \dfrac{N\mu_0 I a^2}{2(r^2 + a^2)^{3/2}}$

Reduces to field at center of loop!

Special Case (r=0): $B = \dfrac{\mu_0 N I a^2}{2a^3} = \dfrac{\mu_0 N I}{2a}$

Ampere's Circuital Law

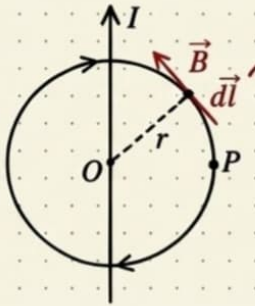
It gives the relationship between the line integral of magnetic field and total current that produces the field.

$$\oint(\vec{B}\cdot\vec{dl}) = \mu_0 \cdot I$$

If $\vec{B}$ is directed towards tangent to every point on perimeter of closed loop:

$$B \cdot L = \mu_0 \cdot I$$

Here, the closed loop is called an Amperian Loop.



Direction of $\vec{B}$ is given by Right Hand Thumb Rule

<u>Proof & Derivation flow</u>

Prove that at any point on a circle/loop, $\vec{B}$ and $\vec{dl}$ are tangential to it.

Let us consider an infinitely long straight wire carrying current I.

We know $\vec{B}$ at distance r is:

$$B = \frac{\mu_0 \cdot 2I}{4\pi \cdot r} = \frac{\mu_0 \cdot I}{2\pi \cdot r}$$

→ Now, $\oint \vec{B}\cdot\vec{dl} = \oint(B \cdot \vec{dl} \cdot \cos(\theta))$

Since tangential, $\theta = 0°$, $\cos(0) = 1$.

$$\oint \vec{B}\cdot\vec{dl} = B \cdot \oint \vec{dl} = B \cdot \ell$$

Substitute B: $\frac{\mu_0 I}{2\pi r} \cdot \ell$

Here length of loop $\ell = 2\pi r$.

$$\boxed{\oint \vec{B}\cdot\vec{dl} = \frac{\mu_0 I}{2\pi r} \cdot 2\pi r = \mu_0 \cdot I}$$

(Proof complete).

Alternative application!

Derive field due to straight wire using Ampere's Law:

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 \cdot I$$

$$B(2\pi r) = \mu_0 \cdot I$$

$$\Rightarrow B = \frac{\mu_0 I}{\ldots}$$

Solenoids & Mediums

Solenoids

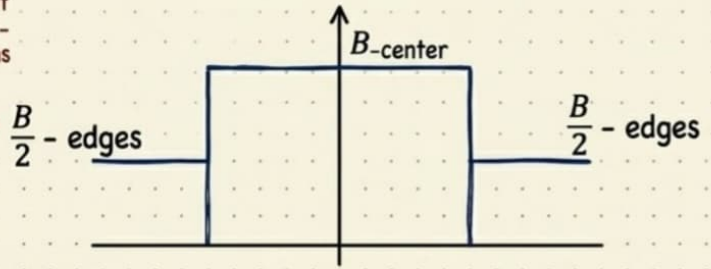
Magnetic field at the edges of the solenoid can be given as:

$$\boxed{B_{edge} = \frac{1}{2}\mu_0 \cdot I \cdot n = \frac{B_{center}}{2}}$$

$$B_{center} = \mu_0 \cdot n \cdot I \text{ (or } \mu_0 \cdot I \cdot N/\ell)$$

Where: n = no. of turns / length, N = total turns, ℓ = length, μ_0 = permeability, I = current.

Expert clarifications



Example Problem: Solenoid with 300 turns/metre, 5A current, length 0.5m, radius 1cm. Find B inside.

Solution: $B = \mu_0 \cdot n \cdot I = (4\pi \cdot 10^{-7}) \cdot 300 \cdot 5$

$= \boxed{6\pi \cdot 10^{-4}\ \text{T}}$ Check units!

Relative Permeability:

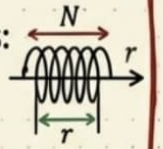
$$B = \mu_0 \cdot n \cdot I$$

If medium is other than air,

$B = \mu_r \cdot B_{air} = \mu_r \cdot \mu_0 \cdot n \cdot I.$

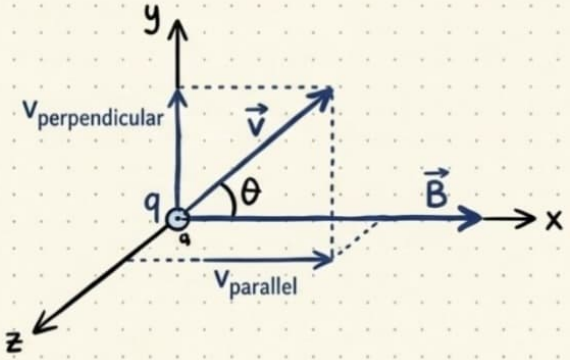
Thus, the field increases! Increases field strength!

Relation between radius & no. of turns: $2\pi \cdot r \cdot N = \ell$. If ℓ is constant, r is proportional to $1/N$.



Part 2: Fields Exert Forces

Force on a Moving Charge in B & Lorentz Force



$F \propto q$ ... (I)

$F \propto v_\perp \Rightarrow v \cdot \sin(\theta)$ ... (II)

$F \propto B$ ... (III)

Combining: $F \propto q \cdot v \cdot B \cdot \sin(\theta) \Rightarrow F = k \cdot q \cdot v \cdot B \cdot \sin(\theta)$

Let $k = 1 \Rightarrow F = q \cdot v \cdot B \cdot \sin(\theta)$

$$\boxed{\text{Vector form: } \vec{F} = q(\vec{v} \times \vec{B})}$$

Dimensional Analysis:

Dimensions of B:

$F = qvB*\sin(\theta) \Rightarrow [M\,L\,T^{-2}] = [L\,T^{-1}] \cdot B \cdot [A\,T]$ (since $q = I*t = [A\,T]$)

$$B = \frac{[M\,L\,T^{-2}]}{[L\,T^{-1}][A\,T]} = [M\,L^0\,T^{-2}\,A^{-1}] = [M\,T^{-2}\,A^{-1}]$$

Final dimension!

Lorentz Force:

A charge q in an electric field (E) experiences force $F_e = qE$.

Lorentz force is the combined total force:
$F = F_e + F_m = qE + q(v \times B) = q[E + (v \times B)]$.

α-particles: He^{2+}.
$q = 2e = 3.2 \cdot 10^{-19}$ C.
$m = 4 \cdot m_p = 6.68 \cdot 10^{-27}$ kg.

Work Done by a Magnetic Field on a Moving Charge is 0

The magnetic field experienced by charge 'q' is: $\vec{F} = q(\vec{v} \times \vec{B})$.

It always acts perpendicular to the velocity $(\vec{v})$ *of the charge.*

Now, $\vec{F} \cdot \vec{v} = 0$

Power $P = \vec{F} \cdot \vec{v}$. *Because they are perpendicular, dot product of perpendicular components is 0.*

Therefore, $q(\vec{v} \times \vec{B}) \cdot \vec{v} = 0$

As per Newton's Second Law: $F = \frac{dp}{dt} = m\left(\frac{dv}{dt}\right)$.

Multiplying both sides by $\vec{v}$:

$$\vec{F} \cdot \vec{v} = m\left(\frac{dv}{dt}\right) \cdot \vec{v} \Rightarrow 0 = m\left(\frac{dv}{dt}\right) \cdot \vec{v}$$

$$0 = \frac{m}{2} * \left[\frac{dv}{dt}(\vec{v} \cdot \vec{v})\right], 0 = \frac{d}{dt}\left[\frac{m}{2}v^2\right], 0 = \frac{d}{dt}\left(\frac{1}{2}mv^2\right)$$

$$0 = \frac{dK}{dt} \Rightarrow \text{Therefore } K = \text{constant.}$$

From Work-Energy theorem, $W = \Delta K = 0$.

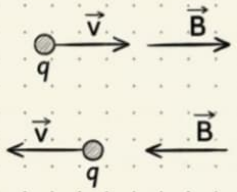
Work done is 0, therefore Energy is constant.

The Particle Trajectory Matrix: Motion in Uniform B

Key Concept:
Only vperpendicular contributes to the magnetic force and circular motion!

Case 1: Parallel / Anti-parallel

Angle: θ = 0 or 180 degrees.



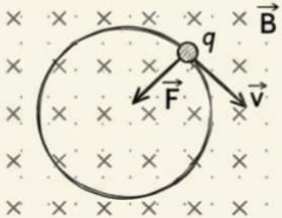
Force: F = qvB*sin(0) = 0.

Path: Straight line.
Charge will go on its original path without any deviation.
Deviation is caused by force (0 in this case).

No force, no deflection!

Case 2: Perpendicular

Angle: θ = 90 degrees.



Force: F = qvB*sin(90) = qvB (max force).

This force acts towards the centre, causing circular motion. qvB = m*v²/R.

$$\text{Radius } R = \frac{m*v}{qB}$$

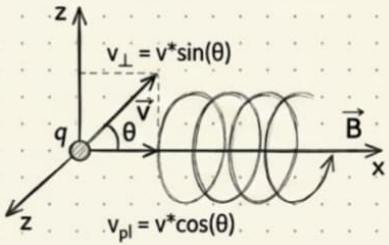
Radius depends on mv (momentum)!

$$\text{Time Period } T = \frac{2*\text{pi}*R}{v} = \frac{2*\text{pi}*m}{qB}$$

$$\text{Frequency } f = \frac{1}{T} = \frac{qB}{2*\text{pi}*m}$$

Case 3: Oblique

Angle: 0 < θ < 90 degrees.



Path: Helical.

$$\text{Radius } r = \frac{m*v_{\text{per},\perp}}{qB} = \frac{m*v*\sin(\theta)}{qB}$$

$$\text{Time Period } T = \frac{2*\text{pi}*m}{qB}$$

$$\text{Pitch} = v_{\text{parallel}} \cdot T = v*\cos(\theta) \cdot \frac{2*\text{pi}*m}{qB}$$

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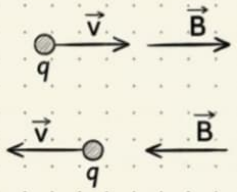

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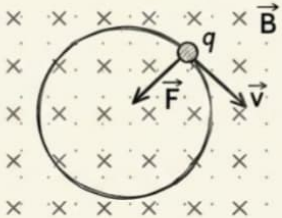

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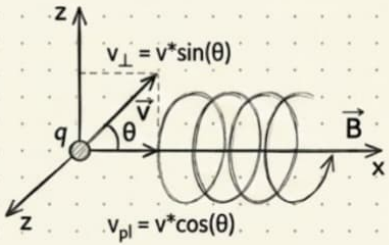


Path: Helical.

$$\text{Radius } r = \frac{m*v_{\text{per},\perp}}{qB} = \frac{m*v*\sin(\theta)}{qB}$$

$$\text{Time Period } T = \frac{2*\text{pi}*m}{qB}$$

$$\text{Pitch} = v_{\text{parallel}} \cdot T = v*\cos(\theta) \cdot \frac{2*\text{pi}*m}{qB}$$

Circular Motion: Energy & Momentum Relationships

We know Radius $r = \frac{m*v}{qB}$.

Also, Kinetic Energy $K.E = \frac{1}{2}mv^2 = qV$ (where V is accelerating potential).

Deriving r from K.E.: $v = \sqrt{2K/m}$

Substitute v into r: $r = m\frac{\sqrt{2K/m}}{qB} = \frac{\sqrt{2mK}}{qB}$

$$\boxed{r = \frac{\sqrt{2mK}}{qB}}$$

Deriving v from potential V: $v = \sqrt{\frac{2K}{m}} \cdot \sqrt{\frac{m}{2qV}} = \sqrt{\frac{2K}{2qV}}$

$$v = \sqrt{\frac{K}{qV}}$$

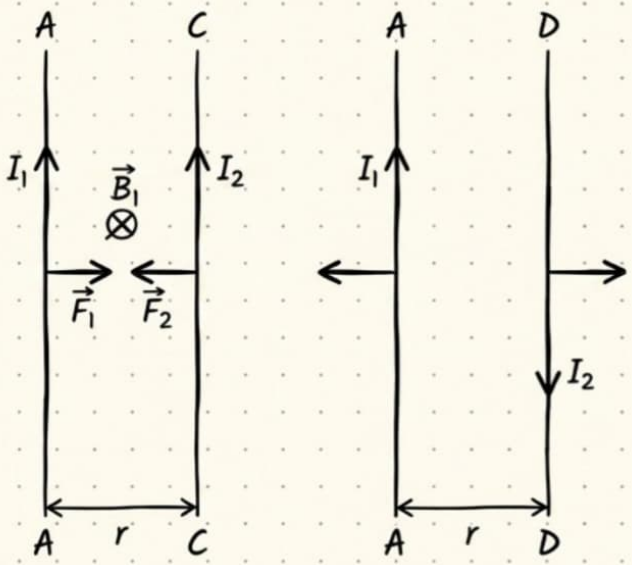
$$\boxed{v = \sqrt{\frac{K}{qV}}}$$

Example:

$$r = \frac{4 \cdot \cancel{9} \cdot \cancel{2}}{(\cancel{4} \cdot 6 \cdot \cancel{2})} \cdot 10^{-31+7+-3+2}$$

$$r = 2.25 \cdot 10^{-6}\,\text{m}$$

Force Between 2 Parallel Current-Carrying Conductors



Let us consider 2 long parallel current carrying conductors AB & CD carrying currents I_1 & I_2 respectively, separated by dist 'r'.

Key Concept: Force is attractive for same direction currents, repulsive for opposite direction currents.

B due to long wire CD at any pt on wire AB:

$$B_2 = \frac{\mu_0 I_2}{2\pi r} \; [\odot]$$

Now, force on 'l' length of wire AB:

$$F_1 = I_1 \ell B_2 \sin(90°)$$

Angle between I_1 and $\vec{B}_2$

$$F_1 = I_1 \ell B_2$$

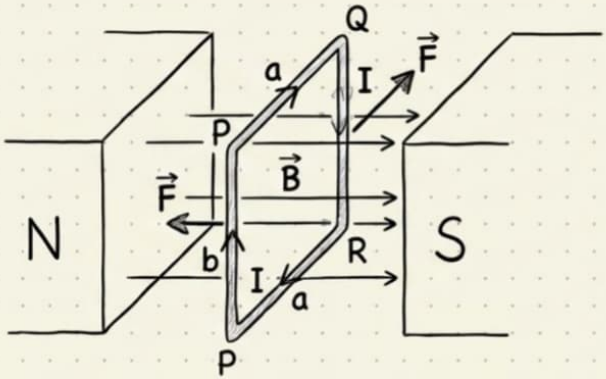
$$F_1 = \frac{I_1 \ell \mu_0 I_2}{2\pi r} = \frac{\mu_0 I_1 I_2 \ell}{2\pi r}$$

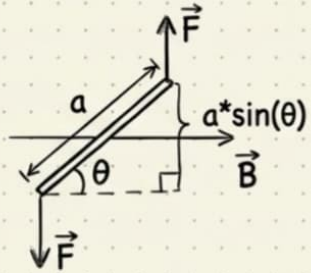
Force acting per unit length:

$$f_1 = \frac{F_1}{\ell} = \frac{\mu_0 I_1 I_2 \ell}{2\pi r \ell}$$

$$\boxed{f_1 = \frac{\mu_0 I_1 I_2}{2\pi r}}$$

Torque Experienced by a Current Carrying Loop





Let us consider a rectangular current carrying loop PQRS suspended in uniform magnetic field B, carrying current I flowing through loop.

Force on arm: $F = I\ell * B * \sin(\theta)$.

Here $\theta = 90° \Rightarrow F = I\ell b * \sin(90) = I\ell * B = Ib * B$ (where l=b).

Torque τ = F x perpendicular distance.

$\tau = I * b * B * a * \sin(\theta) \Rightarrow \tau = I * a * b * B * \sin(\theta)$.

Angle between Area $\vec{V}$ and $\vec{B}$

Since Area $A = a * b$: $\tau = I * A * B * \sin(\theta)$.

For N turns: $\tau = N * I * A * B * \sin(\theta)$.

Magnetic Moment definition: $\vec{m} = N * I * A$.

$$\boxed{\tau = m * B * \sin(\theta) \Rightarrow \vec{\tau} = \vec{m} \times \vec{B}}$$

Key Concept: Torque tends to align the loop with the magnetic field, maximising magnetic flux.

Part 3: Real-World Applications

Moving Coil Galvanometer

Principle: Balances deflecting torque against a restoring spring torque.

Restoring torque: τ_{rest} is proportional to $\alpha \Rightarrow$

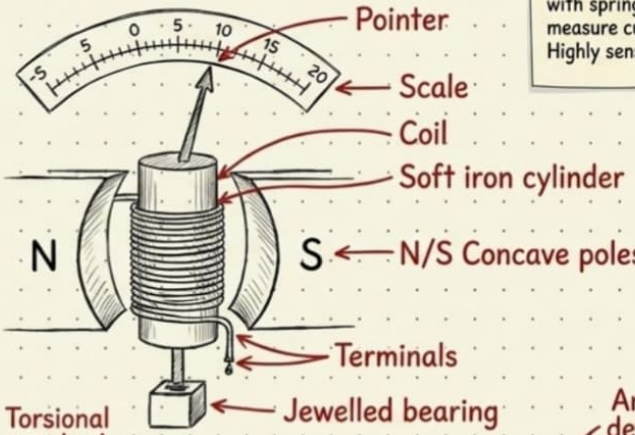
$$\tau_{rest} = k*\alpha$$

(where α is angle of twist, k is torsional constant).

Deflecting torque: $\tau_{def} = N*I*B*A$.

At equilibrium: $k*\alpha = N*I*B*A \Rightarrow I = \frac{k}{N*B*A}*\alpha$.

Figure of Merit (G): $I = G*\alpha \Rightarrow G = \frac{I}{\alpha} = \frac{k}{N*B*A}$.



Key Concept: The galvanometer balances magnetic torque with spring torque to measure current. Highly sensitive.

Torsional constant of spring

Torsional constant of spring

Angle of deflection

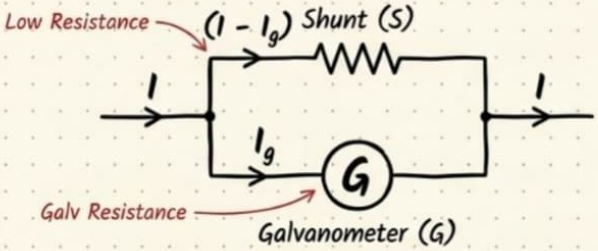
Sensitivities

Current Sensitivity: $I_s = \frac{\alpha}{I} = \frac{N*B*A}{k}$

Voltage Sensitivity: $V_s = \frac{\alpha}{V} = \frac{\alpha}{I*R} = \frac{N*B*A}{k*R}$

Modifying the Galvanometer: The Blueprint

Conversion to Ammeter



Concept: Connect a low resistance Shunt (S) in parallel.

Since G & S are in parallel, Potential Difference (Pd) across G = Pd across S.

$$(I - I_g)*S = I_g * G \Rightarrow \boxed{S = \frac{I_g * G}{I - I_g}}$$

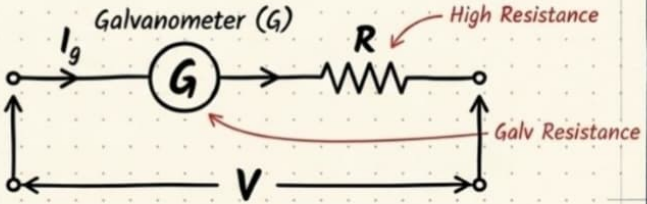
Solving for I_g:

$$(I - I_g)*S = I_g * G \Rightarrow I*S - I_g*S = I_g * G$$

$$I_g * G + I_g * S = I*S \Rightarrow I_g*(G+S) = I*S$$

$$\boxed{I_g = \frac{I*S}{G+S}}$$

Conversion to Voltmeter



Concept: Connect a high resistance resistor (R) in series so very less current passes through the galvanometer. Ideal resistance is infinity.

Total voltage V is sum of voltage drops.

$$V = I_g * (G + R)$$

$$\boxed{I_g = \frac{V}{G + R}}$$

Ideal voltmeter: infinite resistance